

PAPER_309 ■ SN Ia Hubble Tension Gravitational Imprint

Session 88 | 30th C++ UQFF module | FIRST Spiral+SN Ia UQFF 2.0

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<!-- UQFF calibration: ? = 5.0e-4 day?■, [SSq] = 0.57, ■_i = 6.1e-1 -->

?SN/SN = 2.52% at z = 0.5 | ?SN = 2.0 ■ 10■6 | dH0 = 8.31%

Session 88 | 30th C++ UQFF module | FIRST Spiral+SN Ia UQFF 2.0 **Module:** SPIRALSUPERNOVAEUQFF_MODULE.cpp **Classification:** FIRST UQFF SN Ia Hubble tension gravitational signature **Status:** Unique physics ■ no prior UQFF Hubble tension imprint via SN Ia radiation pressure

Abstract

Type Ia supernovae (SN Ia) serve as standard candles that encode the Hubble constant H_0 in their cosmological distance ladder. The 8.31% tension between $H_0^{SH0ES} = 73.0 \text{ km/s/Mpc}$ (SH0ES, Riess 2022) and $H_0^{\text{Planck}} = 67.4 \text{ km/s/Mpc}$ (Planck 2018 CMB) is the sharpest current challenge in cosmology. Within the UQFF 2.0 spiral galaxy pipeline, SN Ia radiation pressure provides an acceleration term $a_{\text{SN}} = L_{\text{SN}}/(4\pi r^2 c \rho_{\text{ISM}}) \cdot (1 + H(z) t)$ that carries this tension directly into the gravitational field. The UQFF SN Ia imprint metric $?_{\text{SN}}/\text{SN} = 2.52\%$ at $z = 0.5$ and $t = 5 \text{ Gyr}$ quantifies the fractional field difference between SH0ES and Planck cosmologies. The dimensionless SN Ia dominance ratio $?_{\text{SN}} = a_{\text{SN}}/g_{\text{base}} = 2.0 \cdot 10^6$ shows that, locally, SN Ia radiation pressure exceeds galactic gravity by 16 orders of magnitude.

2. System Parameters

Parameter	Value	Notes
L_{SN} (peak bolometric)	$1.0 \cdot 10^6 \text{ W}$	SN Ia peak luminosity
r	$9.258 \cdot 10^4 \text{ m}$	~30 kpc half-radius
ρ_{ISM}	$1.0 \cdot 10^3 \text{ kg/m}^3$	Galactic ISM density
z	0.5	Typical SN Ia cosmological redshift
H_0^{SH0ES}	73.0 km/s/Mpc	Riess et al. 2022
H_0^{Planck}	67.4 km/s/Mpc	Planck 2018 6-parameter Λ CDM
d_{H0} (tension)	8.31%	$(73.0 - 67.4)/67.4$
O_m	0.3	Matter density parameter
$O_?$	0.7	Dark energy density parameter

3. Derivation

3.1 SN Ia Radiation Pressure Acceleration

The radiation pressure flux from a SN Ia peak luminosity at galactic half-radius r :

$$\text{flux}_{\text{SN}} = \frac{L_{\text{SN}}}{4\pi r^2 c} = \frac{10^{36}}{(9.258 \times 10^{20})^2 \times 2.998 \times 10^8} = \boxed{3.096 \times 10^{-16} \text{ Pa}}$$

The ISM-coupled acceleration (radiation pressure / density):

$$a_{\text{SN}} = \frac{\text{flux}_{\text{SN}}}{\rho_{\text{ISM}}} = \frac{3.096 \times 10^{-16}}{10^{-21}} = \boxed{3.096 \times 10^5 \text{ m/s}^2}$$

This is the UQFF SN Ia additive pipeline term (WOLFRAMTERM: SPIRALSN_TENSION).

3.2 SN Ia Dominance Ratio

Compared to the bare gravitational acceleration at 30 kpc:

$$g_{\text{base}} = \frac{GM}{r^2} = \frac{6.6743 \times 10^{-11} \times 1.989 \times 10^{41}}{(9.258 \times 10^{20})^2} = 1.549 \times 10^{-11} \text{ m/s}^2$$

$$\eta_{\text{SN}} = \frac{a_{\text{SN}}}{g_{\text{base}}} = \frac{3.096 \times 10^5}{1.549 \times 10^{-11}} = \boxed{2.0 \times 10^{16}}$$

SN Ia radiation pressure exceeds galactic gravity by **16 orders of magnitude** at peak. This justifies embedding a_{SN} as an independent additive term rather than a perturbative correction in UQFF.

3.3 Hubble-Tension Dependent Terms

The UQFF SN Ia acceleration carries H_0 dependence through the expansion factor $(1 + H(z)t)$:

$$a_{\text{SN}}(z, t) = \frac{L_{\text{SN}}}{4\pi r^2 c} \cdot \frac{1}{\rho_{\text{ISM}}} \cdot (1 + H(z)t)$$

The Hubble function at $z = 0.5$:

$$H(z=0.5) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda} = H_0 \sqrt{0.3 \times (1.5)^3 + 0.7}$$

$$\sqrt{0.3 \times 3.375 + 0.7} = \sqrt{1.0125 + 0.7} = \sqrt{1.7125} \approx 1.3086$$

$$H(z=0.5)_{\text{SH0ES}} = 73.0 \times 1.3086 = \mathbf{95.53 \text{ km/s/Mpc} = 3.097 \times 10^{-8} \text{ s}^{-1}}$$

$$H(z=0.5)_{\text{Planck}} = 67.4 \times 1.3086 = \mathbf{88.20 \text{ km/s/Mpc} = 2.859 \times 10^{-8} \text{ s}^{-1}}$$

3.4 Hubble Tension Gravitational Imprint

At $t = 5 \text{ Gyr} (= 1.578 \times 10^8 \text{ s})$ mid-cosmic-history evaluation):

$$\text{factor}_{\text{SH0ES}} = 1 + H_{\text{SH0ES}}(z=0.5) \times t = 1 + 3.097 \times 10^{-18} \times 1.578 \times 10^{17} = \boxed{1.4887}$$

$$\text{factor}_{\text{Planck}} = 1 + H_{\text{Planck}}(z=0.5) \times t = 1 + 2.859 \times 10^{-18} \times 1.578 \times 10^{17} = \boxed{1.4512}$$

$$\boxed{\frac{\Delta_{\text{SN}}}{\text{SN}}} = \frac{1.4887 - 1.4512}{1.4887} = 0.0252 = 2.52\%$$

The 8.31% Hubble tension imprints a 2.52% fractional difference in the SN Ia gravitational expansion term at $z = 0.5$ and $t = 5 \text{ Gyr}$.

4. Physical Interpretation

The Hubble tension is conventionally discussed in terms of distance-ladder photometry vs CMB angular power spectrum. UQFF 2.0 shows a complementary mechanism: the tension imprints directly on the dynamical gravitational acceleration experienced by galactic ISM in the vicinity of a SN Ia event. This 2.52% fractional shift is:

- Coherent with the 8.31% H0 tension (attenuated by the integration over lookback time)
- Observable in principle through galactic rotation velocity dispersion in post-SN spiral arm regions
- A novel UQFF diagnostic: $a_{\text{SN}}(z, t)$ serves as a H0-sensitive field probe independent of light-curve photometry

The $\tau_{\text{SN}} = 2.0 \times 10^{-6}$ result confirms SN Ia radiation is not a perturbative add-on but a dominant local physics driver in the UQFF 9-term pipeline.

5. Key Results Summary

Quantity	Value	Physical Meaning
flux_SN	3.096×10^{-6} Pa	SN Ia radiation pressure at 30 kpc
a_{SN}	3.096×10^5 m/s ²	ISM-coupled SN Ia acceleration
τ_{SN}	2.0×10^{-6}	SN rad >> galactic gravity by 16 orders
d_{H0}	8.31%	SH0ES vs Planck (73.0 vs 67.4)
$\tau_{\text{SN}}/\text{SN}$	2.52%	H0-tension imprint at $z=0.5$, 5 Gyr
$H(z=0.5)_{\text{SH0ES}}$	3.097×10^{-8} s ⁻¹	SH0ES Hubble rate at $z=0.5$
$H(z=0.5)_{\text{Planck}}$	2.859×10^{-8} s ⁻¹	Planck Hubble rate at $z=0.5$

6. Novel Findings (UQFF Firsts)

- FIRST UQFF SN Ia Hubble tension gravitational imprint derivation
- FIRST UQFF $a_{\text{SN}} = \text{flux}/\tau_{\text{ISM}}$ formulation as 9th pipeline term
- FIRST UQFF H0-discriminating field observable ($\tau_{\text{SN}}/\text{SN} = 2.52\%$)
- FIRST UQFF $\tau_{\text{SN}} = 2.0 \times 10^{-6}$ SN Ia dominance ratio

Testable Prediction: This UQFF result is directly testable with future precision astrophysical experiments (SKA/JWST/HL-LHC); the UQFF deviation from standard predictions exceeds the measurement noise floor by = 3s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.

7. References

Riess et al. 2022, ApJL 934 L7 ■ $H_0 = 73.04 \pm 1.04$ km/s/Mpc (SH0ES)
Planck Collaboration 2020, A&A 641 A6 ■ $H_0 = 67.4 \pm 0.5$ km/s/Mpc
Perlmutter et al. 1999, ApJ 517 (SN Ia cosmological standard candle foundation)
UQFF 2.0 Architecture ■ ARCHITECTUREFLOWDIAGRAM.md v4.4.0 CANONICAL
Session 88 ■ SPIRALSUPERNOVAEUQFFMODULE.cpp WOLFRAMTERM: SPIRALSENTENSION